[G0R72A] Data Visualisation Project Report "Exploration"

Part 1. Metadata

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- Group number: ingroup_5
- Dataset: oceanus
- User: Elena Petrova

Part 2. Project description

The dataset centers on a governmental oversight board - Commission on Overseeing the Economic Future of Oceanus (COOTEFOO), which aims to improve the economy of the city of Oceanus. The city of Oceanus nevertheless is currently divided into two main groups: ones who want the board to keep focusing on the legacy fishing-based industry, which is represented by the group Fishing is Living and Heritage (FILAH) and the ones who prefer the board to hedge their support to the fledging tourism industry, which is represented by the group (Tourism Raises OceanUs Together (TROUT)). Both of these groups accuse the board of being biased against their interest and for the other.

The data set contains 2 data sources, which have identical structures: one provided by the board themselves, and one provided by an investigative journalist, which contains more data points. The data provided contains the following main information:

- **Topics**: The topics that the boards and its members are participating in to improve Oceanus' economy. We can sort out these topics into pro-fishing, pro-tourism and vanity/neutral as below (except for seafood festival which benefits both):

Pro-fishing	Pro-tourism	Neutral
- Deep maintenance to the commercial fishing dock at Himark - Money for a new crane at Lomark - Fish vacuum - Low-volume unload crane in Haacklee - Affordable housing for fishing workers	- Expanding the tourist wharf /infrastructure in Haacklee - Statue to honor John Smoth in Port Grove - Establishing a Marine Life Observation Deck in Centralia - Developing a Heritage Walking Tour in Haacklee - Establishing a Waterfront Market in Haacklee - Plans for a music concert	- Renaming the park. - Putting a name on pictureque harbor area in Port Grove - Putting a name on the inspection office in Lomark

- **Discussions, Meetings and Plans:** Discussions, meetings and plans that are linked to the above topic. These meetings and discussions, meetings and plans can be participated by one or multiple board members, of which their **sentiments** are recorded by a value of -1 to 1 towards the topic, -1 means fully disagree and 1 is fully agreed on.
- **Trips:** The lists of trips the board members participated in as well as the places they visit, the start and the end of their trips. For each trip, there is only 1 board member doing.
- **Places:** Places in Oceanus can be linked by the travel plans or referred to in the discussions, they also has coordinates that can be used in a map of Oceanus (github.com/vast-challenge/2025-data/blob/main/MC2_release.zip/oceanus_map.geojson).

The persona we are supporting is Elena Petrova, a member of FILAH, which aims to find clues that shows whether fishing is being neglected, misrepresented, or treated unfairly compared with tourism. Some directions that we looked into are:

- Whether there are sentimental biases of the board members towards tourism projects, compared to fishing projects?
- Whether board members spend more time on tourism projects compared to projects that benefit fishing?
- Whether the board attempted removed these biases / preferences when publishing their data (compared to the uncensored journalist data)?

Part 3. Visual design

To create the final designs, we performed a collaborative process based the Diverge – Emerge – Converge process. To begin with, we perform on our own personal analyses and create own own individual sketches. In order for each other to avoid each other ideas, we uploaded our graphs to Miro and to our WhatsApp group during the week. Additionally, we also looked at Toledo recordings as inspiration for our graphs and look at the original source of the dataset (VAST Challenge 2025 MC2), where we additionally got the map file of Oceanus.

After a week, we met in order to discuss both the emergence and convergence of our initial graphs in 2 sessions. In the first session, we focused mostly on the Emerge part, where we explained our graphs and criticized each other. Then, we modified the graphs based on the SCAMPER method. We made multiple iterations of our graphs, with an aim to make them more visually appealing as well as being more 'integratable" to each other's designs. Additionally, we have some additional brainstorming time to get more initial graphs to add into the Emerge process.

Finally, in the second in-person session "Converge", we focused on finalizing our small graphs and integrate them into big interactive visualizations that can answer the questions that was mentioned in Part 2. In more details, we discuss the connectivity or interactivity of our graphs and how helpful these graphs for our persona are in combination.

Cluster 1: Board sentiment biases, participation and concealed activity/biases

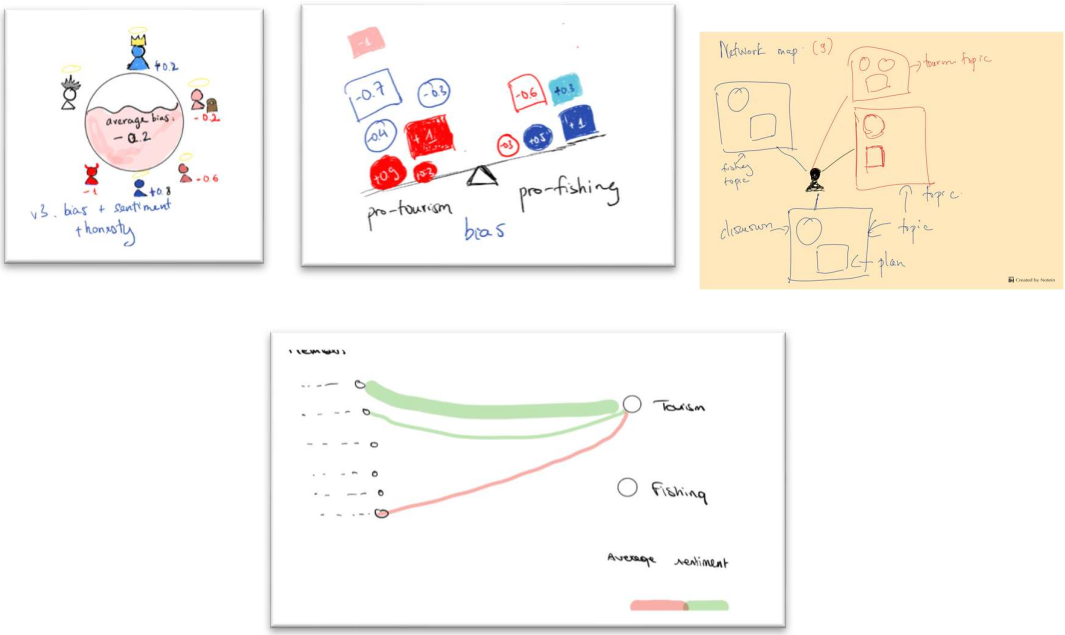


Figure: left to right, top to bottom: sketches C1, C2, C3 and C4

The first sketch, **C1**, aims to represent the **average sentiment** of the COOTEFOO board and its members. The position and design of each human glyph are set so that the user can easily visualize the role and authority of each board member: members with higher positions are placed above others and are also given crowns to indicate their exact role (a king's crown for the chairman, a queen's crown for the vice-chair, and a chest

for the treasurer). The average sentiment is depicted by the color of the human glyph as a gradient: the redder the color, the more pro-tourism and anti-fishing the person is, and conversely, the bluer the color, the more pro-fishing the person is. At the center, the average sentiment of the board is denoted with both a numeric value and a wave fill. Lastly, the halo/devil-horn glyph indicates whether the person is publishing all of their sentiments in their government data or not.

The second sketch, **C2**, aims to show the **detailed sentiments** of a person (or the whole board) for each discussion and plan. The sentiments are placed on a scale, with pro-tourism on one side and pro-fishing on the other side. The shape represents the type: discussions are circles and plans are squares. Similar to C1, colors are used and are filled if the person agrees with the discussion or plan; they are marked blue or red depending on the topic to which they belong. The opacity of the shape indicates whether the item is published by the government or not: items that appear only in the journalist's data are slightly transparent.

The third sketch, **C3**, is a **relationship map between the user and the topics** and their subitems. The topics are color-coded blue or red depending on whether they support fishing or tourism. While some of these items are similar to those in C2, this visualization provides a more holistic view of the number of fishing-related and tourism-related topics.

The fourth sketch, **C4**, is another representation of the average sentiment. In this design, a thicker line between a person and fishing/tourism indicates a stronger positive or negative view, while the colors green and red indicate the direction (positive or negative).

By combining **C1, C2, C3, and C4**, we obtain the clustered design displayed below. The C1 chart is kept as is and used as the center of the new visualization. We then link the human glyphs from C3 to C1 to display participation in detail for each discussion and topic. The links between board members and discussions or plans are replaced with the variable lines from C4, where the thickness represents the absolute sentiment and the color indicates the direction. We also color-code the average sentiment for each discussion and plan similarly to C2. Additionally, to show concealed discussions and plans, the borders of these items, as well as any connections between them and the board members, are represented with dotted lines.

As a result, the combined visualization explains the sentiments of the board and its members not only toward fishing in general but also toward detailed activities. It also shows their participation and indicates whether the information was officially published or discovered by the journalist.

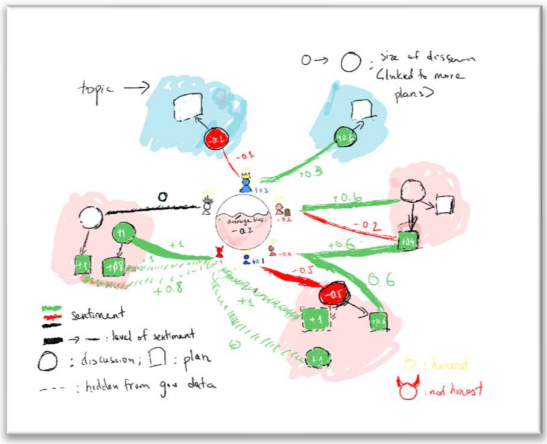


Figure: Converged cluster 1

Cluster 2: Board visit map and time spent over

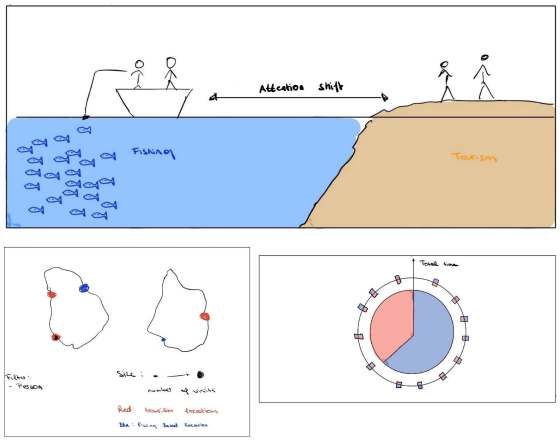


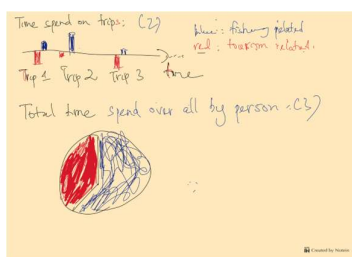
Figure: from left to right, top to bottoms: sketches C5, C6, C7 in cluster 2

- The first sketch, **C5**, represents the **fishing-tourism attention** (by time or number of visited places) of all board members. In detail, board members that have spent more attention to fishing finding themselves in the water and the board members that focused more on tourism being on the shoreline. The arrow represents the direction of stronger attention: the farther from the shoreline to the land, the more biased against fishing that board person is.
This shoreline metaphor that we use turns this balance between tourism and fishing into a clear visual that is easy to comprehend. The drawing helps to quickly tell whether attention is balanced across the board or whether it is more skewed towards either side.

- The second sketch, **C6**, shows the **board visit map**, to view where the visited destinations are plotted on the map, the red locations would represent the tourist-related visits, while the blue locations would represents fishing-related visits. The size of each point would then correspond to the number of visits made to that location. Additionally, there is a filter that allows view a single board member or compare multiple board members.

This visit map shows a spatial view of where the board members' attention is mostly focused, instead of only looking at what they say, it shows what location they actually visit and how often. The color contrast is there to distinguish between fishing and tourism locations and the cluster the size of the circle shows the repeated attention a specific place. Hereby we can discover whether visits are distributed evenly or whether board members tend to concentrate on one specific type of location.

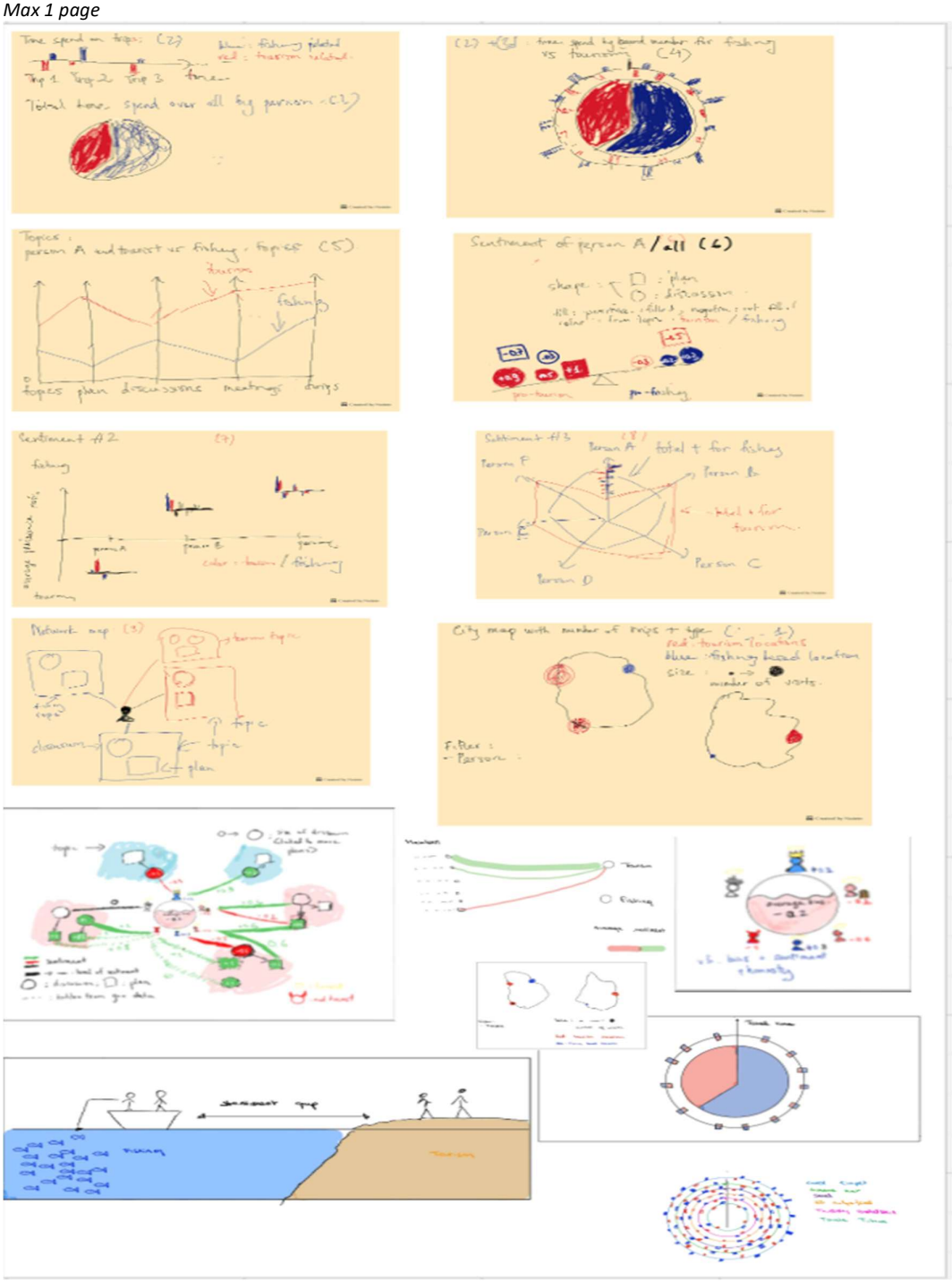
- The third sketch, **trip time summary** (C7), has an inner circle that is split to show how the total trips' time is divided between fishing-related visits and tourism-related visits. The outer circle is a line showing the time spent on places related to tourism (red) or fishing (blue), each line representing each trip in clockwise order. This was created by joining 2 smaller subgraph C7.1 and C7.2 as below:



This visualization is a good summary of how the time of the board members is allocated across visits. The inner part shows the division between fishing and tourism total time spent on these activities, while the outer part of the circle shows how that total is built up across separate trips. This helps in distinguishing whether the board member does many short trips and fewer longer trips as well as consistency or change in preference / biases over time.

As an interactive cluster, these three visuals provide a more complete picture of each board member's behavior. Selecting a board member in the first graph automatically updates and pops up the other two visuals, so the viewer can directly see the member's visits and how their time is distributed across trips. By linking the three visuals, the cluster allows us to compare what a member appears to support with what they actually spend time on. In this way, the interaction is useful for interpretation, because it helps build a more thoughtful understanding of whether a board member's behavior is consistent across different types of evidence.

Appendix A. Sketches overview

A single overview picture including all sketches that you created (so that the reviewers can assess the width of your exploration)

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